Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/10/2005
PI Name:	Johnson, William Ph.D.		
Project Title:	Thermodynamic and Transport Properties of Glass Forming Alloy Liquids Using the Caltech High Vacuum Electrostatic Levitator Platform		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Metals and alloys		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	wlj@caltech.edu	**Fax:**	FY 626-795-6132
PI Organization Type:	UNIVERSITY	**Phone:**	626-395-4433
Organization Name:	California Institute of Technology		
PI Address 1:	Department of Materials Science		
PI Address 2:	Mail Code 138-78		
PI Web Page:	www.its.caltech.edu/~matsci/		
City:	Pasadena	**State:**	CA
Zip Code:	91125	**Congressional District:**	29
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	11/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Kaukler , William	**Contact Phone:**	256-544-0693
Contact Email:	William.Kaukler@msfc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Rhim, Won Kyu (California Institute of Technology)		
Grant/Contract No.:	NNM04AA21G		
Performance Goal No.:			
Performance Goal Text:			

Studies of the thermophysical properties of glass forming liquid metallic alloys are carried out using a High Vacuum Electro-static Levitation System as a platform. The Caltech HVESL system has been developed and upgraded for this purpose. The system utilizes a 4-beam tetrahedral laser heating system to process liquid drops under high vacuum with very uniform drop temperature. We measure liquid heat capacity, viscosity, surface tension, thermal expansion coefficients, and TTT-diagrams for crystallization on glass forming metallic liquids. A variety of methods are used such as high resolution video imaging, drop oscillation studies, and modulation calorimetry. We are developing methods for measurement of atomic diffusion constants in liquid alloys based on the minimization of gravity driven and Marangoni convection. This is accomplished by processing the melt under very near isothermal and quiescent conditions to eliminate gravity driven and Marangoni convection effects on diffusion profiles. We have carried out measurements on a number of glass forming liquid alloys and have developed "process maps" and protocols for carrying out diffusion

Task Description:	measurments. A comprehensive study and analysis of atomic diffusion measurements on the HVES has recently been published [J. Schroers, S. Bossuyt, W.K. Rhim, and W.L. Johnson, Rev. Sci. Instr., 75, 4523 (2004)]. The objective of this task is to carry out accurate measurements of viscosity and atomic diffusion on glass forming liquid metallic alloys both in the equilibrium and undercooled liquid states. The research will evaluate and characterize the effects of gravity on such measurements. Specifically, we wish to evaluate diffusion and viscosity in undercooled liquids as a function of temperature throughout the range from the melting point down to the glass transition. A primary objective is to understand the role of gravity in liquid diffusion studies and develop a "benchmark" micro-gravity experiment which effectively eliminates the influence of convection on diffusion profiles. This task establishes the HVELS platform as the preferred technique for characterizing liquid metals and alloys. The method provides "benchmark" data for viscosity, atomic diffusion, thermal conductivity, surface tension, density, heat capacity, and thermal expansion, on reactive liquids at high temperature. Data on the kinetics of crystallization are used to generate liquid "processing maps" or TTT-diagrams which display the "window of opportunity" for measurements on the undercooled liquid. These data are fundamental inputs for the development of commercial processes for casting and molding of net shape, high performance, amorphous metal hardware and products. The outputs of this project are being directly incorporated into commercial production of amorphous metal products by Liquidmetal Technologies Corp, Ti-Met Corp., and other companies.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Data generated on the Caltech HVESL facility in this work has been directly used for design, simulation, and modeling of manufacturing processes used by Liquidmetal Technologies, Lake Forest, CA to produce metallic glass products. Data for liquid Ti-based alloys has been utilized by Ti-Met Corp., Henderson, NV.
Task Progress:	During the past year, we have carried out measurements of viscosity, thermal expansion, surface tension, heat capacity, and TTT-diagrams for several newly developed bulk metallic glass forming alloys. Alloy systems include: (1) Zr-Co-Cu-Al,(newly developed) (2) Ni-Nb and Ni-Nb-Sn, (3) Zr-Nb-Ni-Cu-Al, (4) Zr-Ti-Ni-Cu-Be., and (5)Pt-Ni-Cu-P (newly developed alloys). The data and analysis of these glass forming liquids is discussed in a number of publications which are listed in the bibliography. We have completed an assessment of the magnitude of convection flows in the liquids during processing. We have determined that gravity driven and Marangoni flows can be significantly reduced in ground based experiments by utilizing our tetrahedral laser heating system to to achieve very near isothermal conditions within levitated drops. We have shown that a micro-gravity experiment will further reduce convective contamination of diffusion profiles and will lead to truly "benchmark" data for atomic diffusion in reactive liquids at high temperatures. We have demonstrated that atomic diffusion measurements over the entire under-cooled liquid region are feasible on alloys of the types (1) and (4) above. We have assessed the time/temperature "window of opportunity" for implementing such measurements. We have demonstrated that a micro-gravity experiment using the present methods will enable the measurement of liquid diffusion with unprecedented accuracy. SUPPLEMENTAL INFORMATION FOR FINAL REPORT: Executive Summary and Project Highlights During the three years of this project, we have made extensive use of the Caltech HVESL factility to develop and utilize benchmark methods for the study of thermo-physical properties of reactive metallic liquids at high temperatures. We completed a study of the HVESL as a platform for carrying out diffusion studies in glass forming liquid metals. The proposed diffusion measurement method utilizes the capabilities of the HVESL to melt and process liquid droplets under highly isothermal and quiescent conditions. We have characterized the low temperature gradients obtained when a liquid metal droplet is melted with a symmetric 4-beam “tetrahedral” laser melting system and demonstrated the temperature variation below 0.2 K are achievable while melting alloys at temperatures between 600 C and 1800 C. These results are documented and reported by Shroers, Rhim,, Zhou, and Johnson in “Enhanced Temperature Uniformity by Tetrahedral Laser Heating “, Rev. of Sci. Instr., 75, 4523 (Nov. 2004). Using experimental data on the temperature gradients and drop conditions, we have modeled gravity driven and Marangoni convection within the drop and shown that internal flow fields can be reduced to the micron/sec range. These results are reported in important paper “Minimizing Convection Effects to Measure Diffusion in Liquid Droplets During High Temperature Electrostatic Levitation” by Bossuyt, Schroers, Rhim, and Johnson, Rev. Sci. Inst. 76, 033909 (2005). Based on the above analysis, we have evaluated the effects of such flows on diffusion in glass forming melts. We have demonstrated that convective contamination of diffusion profiles can be largely eliminated in ground based experiments, and essentially eliminated in a micro-g experiment in low earth orbit. We believe that this analysis and research represents the “state of the art” for studies of atomic diffusion in liquids at elevated temperatures. We have further developed, improved, and implemented are variety of thermo-physical property measurements on the HVESL platform. These include measurements of: 1. Heat Capacity, heat of fusion, and heat of crystallization, of liquid metallic alloy. 2. Viscosity of the liquid alloys using the drop oscillation method. 3. Surface tension of liquid alloys using the drop osciallation method. 4. TTT-diagrams for nucleation and growth of crystals in the undercooled liquids. 5. Volume and Thermal Expansion Coefficients of liquid using high resolution video imaging methods on the HVESL platform. These results on glass forming undercooled liquids have been documented and published in series of scientific papers in the archival literature. Among the results, we have shown a direct correlation between the temperature dependence of the viscosity and crystallization kinetics (TTT-diagrams for crystallization) of glass forming melts [S. Mukherjee et. al., Physical Review Letters.94. 245501 (2005)]. Further, we demonstrated a direct relationship between the molar volume of the liquid and the temperature dependence of the liquid viscosity (liquid fragility) as discussed by Mukherjee et. al. [Phys. Rev. B, 70, 174205 (2004) and Appl. Phys. Lett., 86, 104104 (2005)]. We believe that the Caltech HVESL is currently a “world class” facility for the study of liquids at elevated temperatures. The precision and accuracy of our measurements (in each of the above categories) are second to none. The instrument

	has been used to carry out numerous “benchmark” studies to validate these capabilities. The Caltech HVESL is being actively used as a research tool in the study of development of bulk metallic glass forming alloys. Specifically, we have carried out extensive assessment of the glass forming ability of several Zr-based and Ni-based alloy systems. One particularly important result has been the recognition of the role of liquid viscosity and molar volume of the liquid alloys on glass forming ability and stability of the undercooled melt [S. Mukherjee et. al., Phys. Rev. Lett., 94, 245501 (2005).
Bibliography Type:	Description: (Last Updated: 07/06/2007)
Articles in Peer-reviewed Journals	Bossuyt S, Schroers J, Rhim WK, Johnson WL. "Minimizing convection effects to measure diffusion in liquid droplets during high-temperature electrostatic levitation." Review of Scientific Instruments 2005 Mar;76(3):033909. http://dx.doi.org/10.1063/1.1866932 , Mar-2005
Articles in Peer-reviewed Journals	Mukherjee S, Schroers J, Johnson WL, Rhim WK. "Influence of kinetic and thermodynamic factors on the glass-forming ability of zirconium-based bulk amorphous alloys." Physical Review Letters 2005 Jun 24; 94(24):245501. http://dx.doi.org/10.1103/PhysRevLett.94.245501 , Jun-2005
Articles in Peer-reviewed Journals	Schroers J, Veazey C, Demetriou MD, Johnson WL. "Synthesis method for amorphous metallic foam." Journal of Applied Physics 2004 Dec 15;96(12):7723-30. http://dx.doi.org/10.1063/1.1818355 , Dec-2004
Articles in Peer-reviewed Journals	Schroers J, Johnson WL. "Highly processable bulk metallic glass-forming alloys in the Pt–Co–Ni–Cu–P system." Applied Physics Letters 2004 May 3; 84(18):3666-8. http://dx.doi.org/10.1063/1.1738945 , May-2004
Articles in Peer-reviewed Journals	Schroers J, Lohwangwatana B, Johnson WL, Peker A. "Gold based bulk metallic glass." Applied Physics Letters 2005 Aug 8;87(6):061912. http://dx.doi.org/10.1063/1.2008374 , Aug-2005
Articles in Peer-reviewed Journals	Schroers J, Johnson WL. "Dispersion technique studies on Pd43Ni10Cu27P20." Materials Science and Engineering: A 2004 Jul 15;375-377:781-4. http://dx.doi.org/10.1016/j.msea.2003.10.176 781-784. , Jul-2004
Awards	Johnson WL. "Sigma Mu Lecturer Award, The Metals Society (TMS), Fall meeting, lecture entitled “Potential Energy Landscapes of Metallic Glass Forming Liquids,“ TMS Fall Meeting, Cincinnati, OH, October 2006." Oct-2006
Awards	Johnson WL, Hanan JC, Veazey C. "NASA Tech Brief Award for NTR no 41102, September 2005." Sep-2005